Atoms and their Electrons

1. All the matter in the universe is made of extremely small particles called _____.

2. The charge on a proton exactly balances the electrons _____ _____.

3. _____ discovered that the atom sometimes ejected a much smaller negatively charged particle called an _____.

4. The Thomson model can also be called the _____ _____ model.

5. Heisenberg pictured an atom surrounded by an _____ _____.

6. The smallest atom is _____. It has one _____ in the centre, and one _____ moving quickly around it.

7. _____ has two electrons in the first shell.

8. Lithium has a _____ _____.

9. _____ has four electrons in the outer shell.

10. In Nitrogen, two electrons are paired and there are 3 unpaired in the _____.

11. Fluorine is highly _____ and is the only element to attack _____.

12. Phosphorus has _____ _____ - two paired and three unpaired in the outer shell.

13. All the elements in this group (seven) will try to grab an extra electron but they become more reactive as they get _____.

14. Sharing of electrons is a _____ _____.

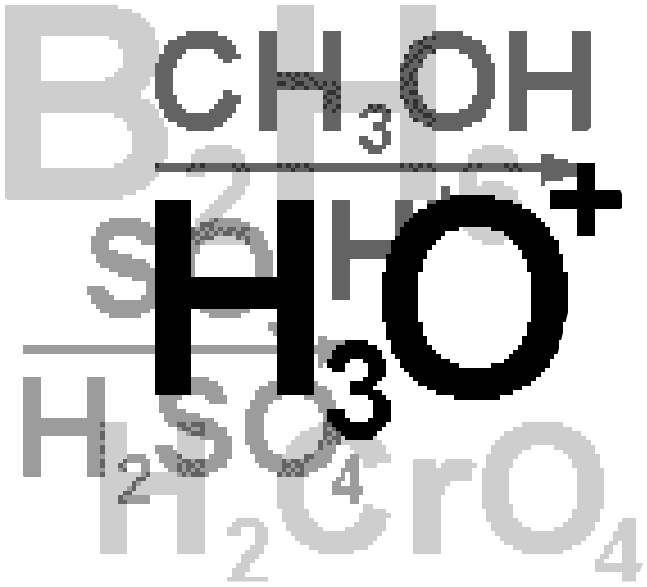
15. Methane is a _____ molecule.

16. Ammonia's shape is similar to _____.

17. _____ has more negative charge on one side of the

molecule making it _____.

18. Carbon dioxide has the charge evenly _____ and the molecule is _____ _____.

19. Each of the atoms has become an _____ and is changed in _____ and _____.

20. The shape of the crystals depends on the ratio of the _____.